

Fine tuning a small LLM with QLoRA for Automated paper review insights

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Abstract

The rapid growth of scientific publications has put unprecedented strain on the traditional peer review system, motivating the exploration of automated solutions which can assist or replace some aspects of review screening process. This study explores the feasibility of utilizing a lightweight large language model, Qwen3-1.7B, for peer review generation through Quantized low rank adaptation (QLoRA), highlighting strengths and weaknesses of a paper. Using the OpenReview 2018 dataset and a set of metrics (ROUGE, BERTScore, Perplexity), the model has been evaluated by comparing two different architectures: baseline model and QLoRa fine tuning, both evaluated with two different prompt strategies (linear and few shot CoT). The results showed that few shot CoT prompt improves the quality of generated reviews compared to a standard linear prompt, while QLoRa outperforms the baseline model in model confidence and linguistic fluency. Despite the promising results, challenges remain in the generative quality, suggesting such lightweight model would better suited for assistance rather than total replacement.
keywords: QLoRa - automated peer reviews - LLM - review generation, CoT

1 Introduction

The peer review process serves as the pillar of scientific publications, ensuring the quality, validity, and rigor of papers through the anonymous evaluation by domain experts.

Beyond these criteria, the feedback from experts can be valuable in the overall improvement of the submitted manuscripts. However the traditional system of manual reviews today faces numerous challenges due to the growth in the number of submissions and the lack of highly specialized reviewers. These challenges resulted in longer review times, increased burden for reviewers and concerns about the sustainability of manual peer review.

Recent advances in large language models (LLMs) have opened new possibilities for addressing these scalability challenges. The demonstrated capabilities of LLMs in understanding complex text, generating coherent responses, and performing domain specific reasoning have the potential of partially or fully automating aspects of the review process.

The most recent works addressed the need of automating this process by exploring resource efficient approaches to automated review assistance.

Source code: <https://github.com/SamC-dev/QLoRA-for-Automated-Paper-Review-Insights>

2 State of the art

2.1 literature review

The exponential growth of scientific publications has created unprecedented challenges for traditional peer review systems. As Yuan et al. (2021) note, the rapid development of science and technology has been accompanied by a volume of submissions that outpaces the availability of qualified human experts. This volume growth, combined with the inherent "poor efficiency and low reproducibility" of traditional peer review (Lin et al., 2023), has motivated the shift towards automated solutions.

Foundational work in this area identified that the peer review system is not only burdened by volume, but systemic flaws, including lacks of reproducibility caused by a subjective component during the evaluation, and the cognitive limitations of reviewers, often specialized in specific subfields. While initial works found out that machines could have definitely been used as assistants during the process, soon studies experimented on making the machine more independent.

Yuan et al. (2021) provided the foundations of this transition, proposing a NLP model capable of generating reviews for scientific papers. The system was able to generate more detailed reviews, offering a starting point in this direction. This shift was formal-

ized by Lin et al. (2023), who introduced the concept of Automated Scholarly Paper Review (ASPR), developing a comprehensive "pipeline... for achieving a full-scale computerized reviewing process." They outlined the biases brought by human reviewers while also highlighting the challenges of ASPRs: imperfect document parsing and representation, defective human-computer interaction, and flawed deep logical reasoning.

Recent advances demonstrated remarkable progress in addressing these limitations. Wu et al. (2025) demonstrated that LLMs could match or exceed human level quality in certain evaluation facets by using a four module architecture: literature search, topic formulation, knowledge extraction, and review composition, supported by multi-layered quality control to reduce the risk of hallucination.

The 2025 survey by Lin et al. provided a holistic view of ASPR in the current LLM landscape, examining which models are being used for automated review and identifying new methods, new datasets, new source codes, and new online systems that come with LLMs for ASPR.

The evolution from AI assisted tools to fully automated ones is clear. The progression of this research establishes a foundation for fine tuning approaches. This study builds upon this foundation by exploring if QLoRA can be used in combination with a lightweight model to address the aforementioned challenges.

2.2 Research question

The central objective of this study is to determine the efficacy of lightweight models in specialized academic reasoning tasks. Specifically, the study seeks to answer the following:

- RQ1: Can a lightweight model (1.7B) effectively generate structured peer reviews, and what is the role of prompt design in the improvement of reviews quality?
- RQ2: To what extent does QLoRA fine tuning improve the overall quality, semantic alignment, and technical consistency of generated reviews compared to the baseline model?

3 Research methodology

3.1 Data exploration and cleaning

This study utilizes the OpenReview 2018 dataset, comprising around 4700 reviews for a total of around 1500 unique papers. From the original dataset only

two columns have been used: abstract and review, which has been cleaned in their string formatting. All data has been furthermore cleaned of missing values by removing rows containing null values.

The original dataset had on average 3 reviews for each unique paper, in order to reduce the overall size and training time, only the subset of all unique papers has been taken, filtering for the longest review in order to give the model the most complete and vast information possible while also reducing the dataset size. The resulting cleaned dataset comprised around 1500 entries.

The dataset has been split into train, validation and test sets according to the following proportion: (8:1:1), this one guaranteed that the training set size was big and varied enough for a small model such as the one employed in this analysis.

To each review was associated a rating to the paper from 1 (very poor) to 10 (excellent).

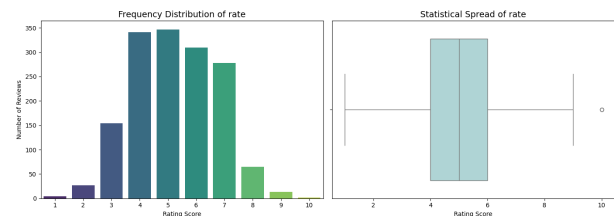


Figure 1: Rating distribution

Despite these ratings being included in the descriptive statistics, they were intentionally excluded from the model generation task. This decision was made to focus the research exclusively on the model’s ability to synthesize high level qualitative critiques, decoupling text generation from numerical assignments, this way avoiding the complexities of numerical predictions by a language model.

3.2 Model

The study employed Qwen 3-1.7B as baseline architecture. To adapt this general purpose model to the specialized domain of scientific peer reviews QLoRa (quantized low rank adaptation) have been used, the quantized version of LoRa.

The baseline model comprises of around 1.7 billion parameters, and fine tuning would normally require to update all of them, which is computationally prohibitive for most settings. LoRa bypass this by freezing the original pre trained weight matrices and injecting trainable low rank decomposition matrices into the transformer layers. During training, only these small matrices are updated. The final out-

put is represented as:

$$h = Wx + BAx$$

Where W is the original frozen weight matrix and A and B are the injected ones. The product BA represents the learned task specific adaptation.

The quantized version, QLoRa, allows the model to be trained on even less performing hardware by introducing the following elements:

- Weights compression from 16 bits to 4 bits without a significant loss in precision.
- Double quantization, allowing a further reduction in the memory footprint.
- Paged optimizers, a feature which manages memory spikes, reducing the risk of hitting OOM (out of memory) errors during training.

3.3 Formalization of the experiments

The experiments followed a comparative approach, taking into consideration QLoRa fine tuning with respect to ground truth and the model’s zero shot capabilities. The evaluation focused on the quality of reviews, assessed using the following metrics:

- ROUGE: Measures the overlap between the generated reviews and human authored reference reviews, It serves as a proxy for content coverage and lexical similarity. Three variants have been covered: ROUGE1 (unigram overlap), ROUGE2 (bigram overlap) and ROUGEL (longest common subsequence).
- BERTscore (RoBERTa as backbone): Computes semantic similarity between generated and reference reviews using contextual embeddings from a pre trained model, providing a more meaning aware assessment than simple n-gram overlap.
- Perplexity: Evaluates how well the language model predicts the next word in the reference text, it serves as a measure of model confidence and linguistic fluency. Values are distributed in the range 1-100, with lower ones indicating a more fluent and consistent text. Formally, for a sequence $X = (x_1, x_2, \dots, x_t)$ it is defined as

$$PPL(X) = \exp \left(-\frac{1}{t} \sum_{i=1}^t \log p(x_i | x_{<i}) \right)$$

3.4 Prompt design

The performance of lightweight LLMs is highly sensitive to the structure and content of the system prompt. To test this dependency in the context of this analysis two different prompt designs were implemented and compared: a standard linear one and a few shot CoT (Chain of thought) architecture:

- standard linear: a top down hierarchy composed by role, instructions and expected formatted output.
- Few shot CoT: combines in context learning and intermediate reasoning. The instructions were sided by a reasoning flow the model had to follow while completing them, furthermore having at disposal an example of expected behavior (example abstract, review and expected reasoning step), this way the model is asked to follow some intermediate steps before giving its final answer, reducing the risk of hallucinating and increasing its logical consistency.

3.5 Training

Training has been done using the following setting:

- Batch size set to 4, with gradient accumulation. Trials with larger batch sizes have been proven unsuccessful due to memory constraints.
- ADAM as the optimizer of choice, it’s the one used in the original paper, it represents the state of the art in terms of optimization algorithms and it typically performs better than other conventional optimizers such as SGD.
- Number of epochs set to 10 in order to find a compromise between resource constraints and optimal results.
- Hyperparameters set in accordance to the original implementation.
- An early stop criterion, based on steps and with patience equal to 2, has been implemented in order to stop training the moment the validation loss didn’t drop further.

4 Results and Discussion

By visually comparing a sample of generated reviews with their ground truth, it’s evident that the model still requires refinement. Nonetheless the reviews capture the strengths and weaknesses of their papers

quite effectively. The following table summarizes the Rouge and BERT scores of the baseline model by using the two different prompt structures (linear and few shot CoT):

	Metric	Standard Prompt	Few-Shot CoT	Delta (%)
0	Perplexity	16.783026	22.995779	37.02
1	ROUGE-1 (F1)	0.208977	0.262790	25.75
2	ROUGE-2 (F1)	0.040011	0.053459	33.61
3	ROUGE-L (F1)	0.121062	0.137020	13.18
4	BERTScore (F1)	0.798887	0.821527	2.83

Figure 2: Comparison of results: standard prompt vs. few shot CoT

in the few shot setting, CoT performed slightly better than the standard prompt across all metrics, with the curious exception of perplexity. Few shot CoT resulted in a increase of about 37% in perplexity, possibly due to different reasons:

- By introducing a reasoning intermediate step and forcing the model to set a thinking trace, the number of decision points the model had to take increased.
- The model transitioned from a direct linear process to a layered one, increasing the overall uncertainty.
- mismatch between reasoning task and ground truth reference reviews: There is a level of "surprise" by the model discovering that the human reviews didn't follow its own internal reasoning.

Even if the model was less sure of the next word of the generated sequence, the results in the other metrics clearly showed an improvement, proving a well designed prompt has to be favored.

The fine tuned model have also been evaluated with the two different prompts:

The standard prompt resulted in overall better performance, with the exception of perplexity, this one extremely low using the few shot CoT prompt.

Surprisingly enough, there is not much of a difference between using QLoRa and the baseline model in terms of performance, with actually a slight drop in favor of the baseline one, exception made again for perplexity, where QLoRa in combination with few shot CoT prompt achieved the best result. Furthermore the CoT prompt in the fine tuned setting performed worst than the standard one. These results may have been caused by few architectural factors:

	Metric	QLoRA_Std	QLoRA_CoT	Delta %
0	Perplexity	17.7530	7.2031	-59.4262
1	ROUGE-1	0.2570	0.2084	-18.9002
2	ROUGE-2	0.0452	0.0359	-20.6159
3	ROUGE-L	0.1320	0.1086	-17.7018
4	BERTScore	0.8118	0.7947	-2.1040

Figure 3: Comparison of results: QLoRa standard prompt vs. few shot CoT

- The QLoRa configuration prioritized memory efficiency over parameter rich adaptation.
- Only one combination of model parameters was used, with no ablation studies to choose the best one.
- CoT design may have influenced the final outcome.

5 Conclusion

This study explored the potential of fine tuning a lightweight LLM using QLoRa for automated scholarly paper review.

A primary finding of this research is that while QLoRa led to a slight decrease in lexical overlap and semantic similarity metrics compared to the baseline, it achieved a significant improvement in model confidence and linguistic stability, with Perplexity dropping to 7.2, indicating that the model successfully internalized the structural and stylistic conventions of peer review domain.

Experimenting with different prompting strategies revealed successful, with few shot CoT performing overall better in the baseline model setting and giving useful hints in a fine tuned setting.

It can be concluded that a baseline model is capable of assisting in the process of review generation and that QLoRa fine tuning only partially improves its performance.

Despite these encouraging results, future work should address several limitations:

- The study relies on the OpenReview 2018 dataset, which, while substantial, may not fully capture the diversity of scientific domains or review styles.

- The training process was limited due to resource constraints, more sophisticated and structured ablation studies are needed.
- Different lightweight models should be tested.

6 References

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